

SARA ALI MAHRAN

Senior Software Engineer

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SUMMARY

Senior Software Engineer with 3+ years of experience building scalable Python systems. Proven in backend optimization, modular architecture migrations, and delivering production-grade solutions — with 99.4%+ uptime to show for it. Strong background in mentoring technical teams and adapting complex concepts for diverse learners.

RELEVANT EXPERIENCE

Senior Software Engineer | Planning Engineer Company

Apr 2025 – Present

Tech: Python, PySide6, FastAPI, Django, PostgreSQL, PgBouncer, SQLite, Redis, SSE, Docker, nginx, Linux

- Designed and developed cross-platform desktop applications for planning engineers using Python, PySide6, and PostgreSQL, automating scheduling, data visualization, and Oracle Primavera P6 integration through an XER import/export engine built from scratch.
- Architected multi-tenant data isolation by provisioning a dedicated managed PostgreSQL database per client organization, with PgBouncer connection pooling, encrypted credentials, and dynamic per-tenant routing.
- Led a full audit and hardening of session and token security: refresh token rotation with reuse detection, OS keyring token storage, login rate limiting, device-based session control, and a role model separating internal staff from client users.
- Built real-time event delivery over Server-Sent Events (SSE), powering an in-app notification system, instant remote logout, and live subscription expiry enforcement across the client and admin applications.
- Designed the subscription and trial lifecycle end to end, covering entitlement enforcement, renewal flows, trial feature gating, and offline-safe expiry checks.
- Boosted system reliability to 99.4%+ uptime and led the migration from a monolith to modular, independently deployable tools, improving scalability and maintainability.
- Own production infrastructure on DigitalOcean end to end: Dockerized FastAPI services orchestrated with Docker Compose, nginx, managed PostgreSQL, automated backups, and health alerting.
- Conducted 31 candidate interviews: 26 screening interviews assessing problem solving and technical mindset, and 5 live coding interviews evaluating engineering reasoning.

Session Lead | Udacity

Jul 2024 – Sep 2025

- Delivered 40+ technical sessions in Python, React, and React Native to 100+ youth and adult learners across DECI and DCDE programs, adapting content for absolute beginners through advanced builders.
- Achieved learner graduation rates of 86–96% across 5 cohorts — consistently at or above the top performance threshold — through structured mentorship and adaptive teaching.
- Mentored 130+ learners through real-world projects, from debugging to clean code practices — several have since landed junior developer roles.

TECHNICAL SKILLS

Languages: Python (PySide6/Qt), JavaScript, TypeScript

Backend & APIs: FastAPI, Django, REST APIs, Server-Sent Events (SSE), JWT, OAuth, Session & Token Security

Databases: PostgreSQL, PgBouncer, SQLite, Redis, SQLAlchemy

Architecture: System Design, Multi-Tenant Architecture, Modular Architecture, Monolith → Modular Services

Infrastructure: Docker, Docker Compose, nginx, DigitalOcean, API Design & Service Integration

Tools: Git, GitHub, Linux

EDUCATION

Bachelor's Degree, Software Engineering

Oct 2016 – May 2022

Cairo University